MSG No. 132.457 P_C4_2/mcm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `m.mm`

No.	position	mapping
1	$[0, 0, 0]$	`[1,6,7,8,9,14,15,16]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,3,4,5,10,11,12,13]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[17,22,23,24,25,30,31,32]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[18,19,20,21,26,27,28,29]`

Table 2: Wyckoff site: `4b`, site symmetry: `m.m'm'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,6,9,14,23,24,31,32]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,10,11,20,21,28,29]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,5,12,13,18,19,26,27]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[7,8,15,16,17,22,25,30]`

Table 3: Wyckoff site: `4c`, site symmetry: `m.mm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,8,14,15,22,23,25,32]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,4,11,13,19,21,26,28]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,5,10,12,18,20,27,29]`
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[6,7,9,16,17,24,30,31]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.mm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,8,14,15,22,23,25,32]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,4,11,13,19,21,26,28]`
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[3,5,10,12,18,20,27,29]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,9,16,17,24,30,31]`

Table 5: Wyckoff site: `4e`, site symmetry: `-42m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,5,6,10,11,15,16]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,7,8,9,12,13,14]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,20,21,22,26,27,31,32]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[18,19,23,24,25,28,29,30]`

Table 6: Wyckoff site: `4f`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,5,6,26,27,31,32]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[2,3,7,8,25,28,29,30]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,12,13,14,18,19,23,24]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[10,11,15,16,17,20,21,22]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6,15,16]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,3,12,13]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,10,11]`
4	$[0,\ 0,\ -z]$	`[7,8,9,14]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[17,22,31,32]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[18,19,28,29]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[20,21,26,27]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[23,24,25,30]`

Table 8: Wyckoff site: `8h`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,31,32]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,28,29]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,5,26,27]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8,25,30]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,14,23,24]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[10,11,20,21]`
7	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,13,18,19]`
8	$\left[\frac{1}{2},\ 0,\ z\right]$	`[15,16,17,22]`

Table 9: Wyckoff site: `8i`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,15,22,32]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,13,19,28]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3,12,18,29]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[4,11,21,26]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[5,10,20,27]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[6,16,17,31]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[7,9,24,30]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[8,14,23,25]`

Table 10: Wyckoff site: `8j`, site symmetry: `m.2m`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7,14,16]`
2	$[-x,\ x,\ \frac{1}{2}]$	`[2,5,11,12]`
3	$[x,\ -x,\ \frac{1}{2}]$	`[3,4,10,13]`
4	$[-x,\ -x,\ 0]$	`[6,8,9,15]`
5	$[x+\frac{1}{2},\ x+\frac{1}{2},\ 0]$	`[17,23,30,32]`
6	$[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}]$	`[18,21,27,28]`
7	$[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[19,20,26,29]`
8	$[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0]$	`[22,24,25,31]`

Table 11: Wyckoff site: `8k`, site symmetry: `m.2'm'`

No.	position	mapping
1	$[x,\ x+\frac{1}{2},\ \frac{1}{2}]$	`[1,14,23,32]`
2	$[\frac{1}{2}-x,\ x,\ 0]$	`[2,11,21,28]`
3	$[x+\frac{1}{2},\ -x,\ 0]$	`[3,10,20,29]`
4	$[x,\ \frac{1}{2}-x,\ 0]$	`[4,13,19,26]`
5	$[-x,\ x+\frac{1}{2},\ 0]$	`[5,12,18,27]`
6	$[-x,\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[6,9,24,31]`
7	$[x+\frac{1}{2},\ x,\ \frac{1}{2}]$	`[7,16,17,30]`
8	$[\frac{1}{2}-x,\ -x,\ \frac{1}{2}]$	`[8,15,22,25]`

Table 12: Wyckoff site: `8l`, site symmetry: `m.2m`

No.	position	mapping
1	$[x,\ x,\ \frac{1}{2}]$	`[1,7,14,16]`
2	$[-x,\ x,\ 0]$	`[2,5,11,12]`
3	$[x,\ -x,\ 0]$	`[3,4,10,13]`
4	$[-x,\ -x,\ \frac{1}{2}]$	`[6,8,9,15]`
5	$[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}]$	`[17,23,30,32]`
6	$[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0]$	`[18,21,27,28]`
7	$[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0]$	`[19,20,26,29]`
8	$[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[22,24,25,31]`

Table 13: Wyckoff site: `8m`, site symmetry: `m.2'm'`

No.	position	mapping
1	$[x,\ x+\frac{1}{2},\ 0]$	`[1,14,23,32]`
2	$[\frac{1}{2}-x,\ x,\ \frac{1}{2}]$	`[2,11,21,28]`
3	$[x+\frac{1}{2},\ -x,\ \frac{1}{2}]$	`[3,10,20,29]`
4	$[x,\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[4,13,19,26]`
5	$[-x,\ x+\frac{1}{2},\ \frac{1}{2}]$	`[5,12,18,27]`
6	$[-x,\ \frac{1}{2}-x,\ 0]$	`[6,9,24,31]`
7	$[x+\frac{1}{2},\ x,\ 0]$	`[7,16,17,30]`

continued ...

Table 13

No.	position	mapping
8	$\left[\frac{1}{2}-x,\,-x,\,0\right]$	`[8,15,22,25]`

Table 14: Wyckoff site: `16n`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0,\,y,\,\frac{1}{4}\right]$	`[1,5]`
2	$\left[-y,\,0,\,\frac{3}{4}\right]$	`[2,8]`
3	$\left[y,\,0,\,\frac{3}{4}\right]$	`[3,7]`
4	$\left[0,\,-y,\,\frac{1}{4}\right]$	`[4,6]`
5	$\left[0,\,-y,\,\frac{3}{4}\right]$	`[9,13]`
6	$\left[y,\,0,\,\frac{1}{4}\right]$	`[10,16]`
7	$\left[-y,\,0,\,\frac{1}{4}\right]$	`[11,15]`
8	$\left[0,\,y,\,\frac{3}{4}\right]$	`[12,14]`
9	$\left[\frac{1}{2},\,y+\frac{1}{2},\,\frac{1}{4}\right]$	`[17,21]`
10	$\left[\frac{1}{2}-y,\,\frac{1}{2},\,\frac{3}{4}\right]$	`[18,24]`
11	$\left[y+\frac{1}{2},\,\frac{1}{2},\,\frac{3}{4}\right]$	`[19,23]`
12	$\left[\frac{1}{2},\,\frac{1}{2}-y,\,\frac{1}{4}\right]$	`[20,22]`
13	$\left[\frac{1}{2},\,\frac{1}{2}-y,\,\frac{3}{4}\right]$	`[25,29]`
14	$\left[y+\frac{1}{2},\,\frac{1}{2},\,\frac{1}{4}\right]$	`[26,32]`
15	$\left[\frac{1}{2}-y,\,\frac{1}{2},\,\frac{1}{4}\right]$	`[27,31]`
16	$\left[\frac{1}{2},\,y+\frac{1}{2},\,\frac{3}{4}\right]$	`[28,30]`

Table 15: Wyckoff site: `16o`, site symmetry: `..m`

No.	position	mapping
1	$[x,\,-x,\,z]$	`[1,15]`
2	$\left[x,\,x,\,z+\frac{1}{2}\right]$	`[2,13]`
3	$\left[-x,\,-x,\,z+\frac{1}{2}\right]$	`[3,12]`
4	$\left[x,\,x,\,\frac{1}{2}-z\right]$	`[4,11]`
5	$\left[-x,\,-x,\,\frac{1}{2}-z\right]$	`[5,10]`
6	$[-x,\,x,\,z]$	`[6,16]`
7	$[-x,\,x,\,-z]$	`[7,9]`
8	$[x,\,-x,\,-z]$	`[8,14]`
9	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,z\right]$	`[17,31]`
10	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$	`[18,29]`
11	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$	`[19,28]`
12	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}-z\right]$	`[20,27]`
13	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,\frac{1}{2}-z\right]$	`[21,26]`
14	$\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,z\right]$	`[22,32]`
15	$\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,-z\right]$	`[23,25]`
16	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,-z\right]$	`[24,30]`

Table 16: Wyckoff site: `16p`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ \frac{1}{2}-x,\ z\right]$	`[1,31]`
2	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	`[2,29]`
3	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	`[3,28]`
4	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,27]`
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[5,26]`
6	$\left[-x,\ x+\frac{1}{2},\ z\right]$	`[6,32]`
7	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	`[7,25]`
8	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	`[8,30]`
9	$\left[-x,\ x+\frac{1}{2},\ -z\right]$	`[9,23]`
10	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	`[10,21]`
11	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	`[11,20]`
12	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[12,19]`
13	$\left[x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[13,18]`
14	$\left[x,\ \frac{1}{2}-x,\ -z\right]$	`[14,24]`
15	$\left[x+\frac{1}{2},\ -x,\ z\right]$	`[15,17]`
16	$\left[\frac{1}{2}-x,\ x,\ z\right]$	`[16,22]`

Table 17: Wyckoff site: `16q`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[2,11]`
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[3,10]`
4	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[4,13]`
5	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[5,12]`
6	$[-x,\ -y,\ 0]$	`[6,9]`
7	$[y,\ x,\ 0]$	`[7,16]`
8	$[-y,\ -x,\ 0]$	`[8,15]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[17,30]`
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[18,27]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[19,26]`
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[20,29]`
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[21,28]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$	`[22,25]`
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[23,32]`
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	`[24,31]`

Table 18: Wyckoff site: `32r`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[4]`

continued ...

Table 18

No.	position	mapping
5	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[5]
6	$[-x,\, -y,\, z]$	[6]
7	$[y,\, x,\, -z]$	[7]
8	$[-y,\, -x,\, -z]$	[8]
9	$[-x,\, -y,\, -z]$	[9]
10	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[10]
11	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$[-y,\, -x,\, z]$	[15]
16	$[y,\, x,\, z]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[22]
23	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[23]
24	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[28]
29	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]